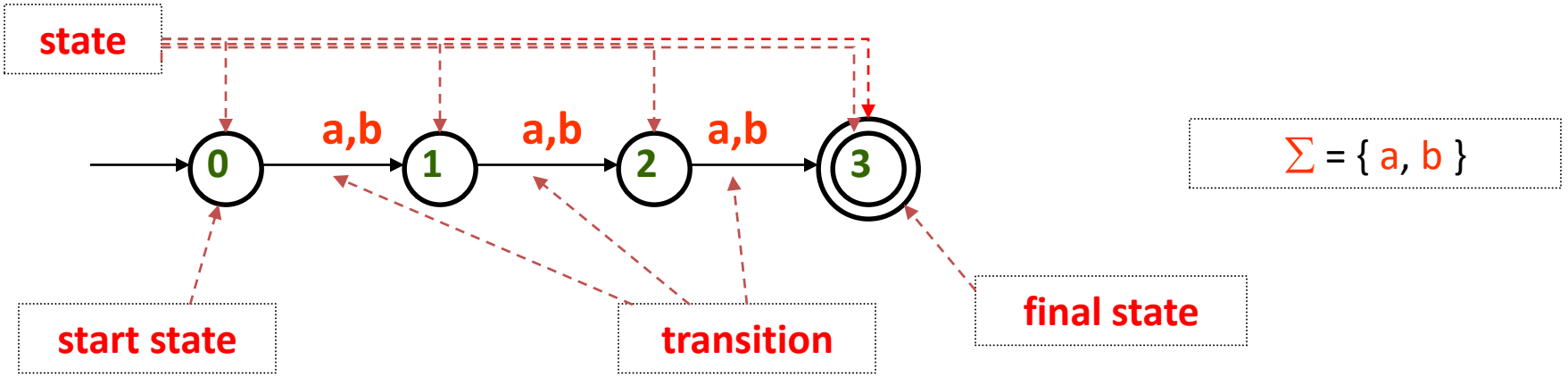
Deterministic Finite Automata

- Also known as Finite state machine, Finite state automata
- It represents an abstract machine which is used to represent a regular language
- A regular expression can also be represented using Finite Automata
- There are two ways to specify an FA: Transition Tables and Directed Graphs.

Graph Representation

- Each node (or vertex) represents a state, and the edges (or arcs) connecting the nodes represent the corresponding transitions.
- Each state can be labeled.

Finite Automata



- Table Representation
  (continued)
  - An FSA may also be represented with a state-transition table. The table for the above FSA:

State	Input a	Input b
0	1	1
1	2	2
2	3	3

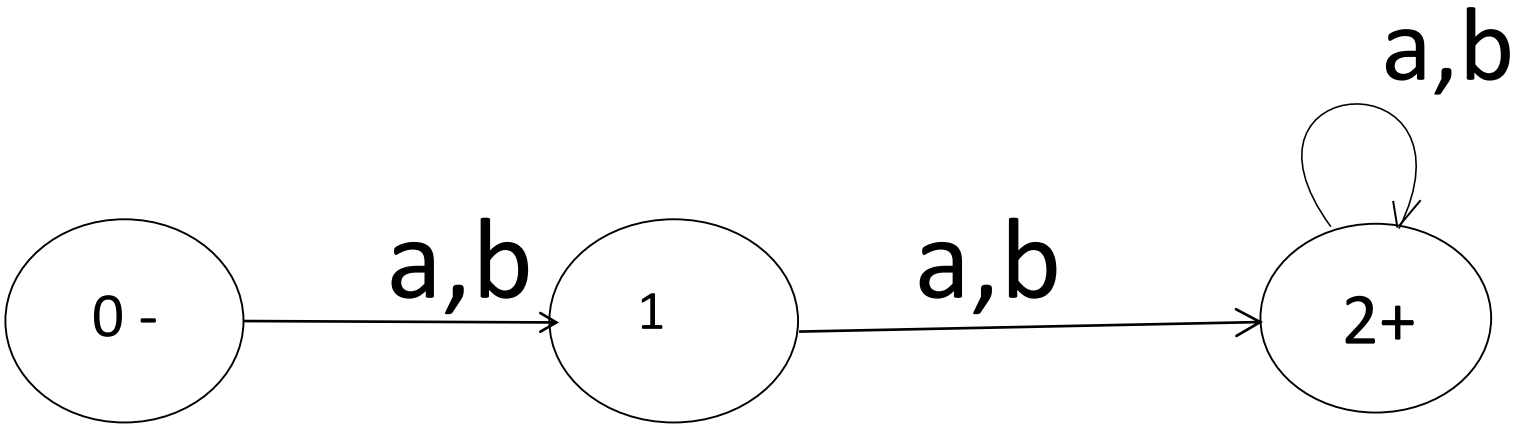
Finite Automata Definition

- An FA is defined as follows:-
  - Finite no of states in which one state must be initial state and more than one or may be none can be the final states.
  - Sigma Σ provides the input letters from which input strings can be formed.
  - **FA Distinguishing Rule**: For each state, there must be an out going transition for each input letter in Sigma Σ.

Another way of representation…

Old States	New States	
	Reading a	Reading b
0 -	1	1
1	2	2
2 +	2	2

FA directed graph



Example

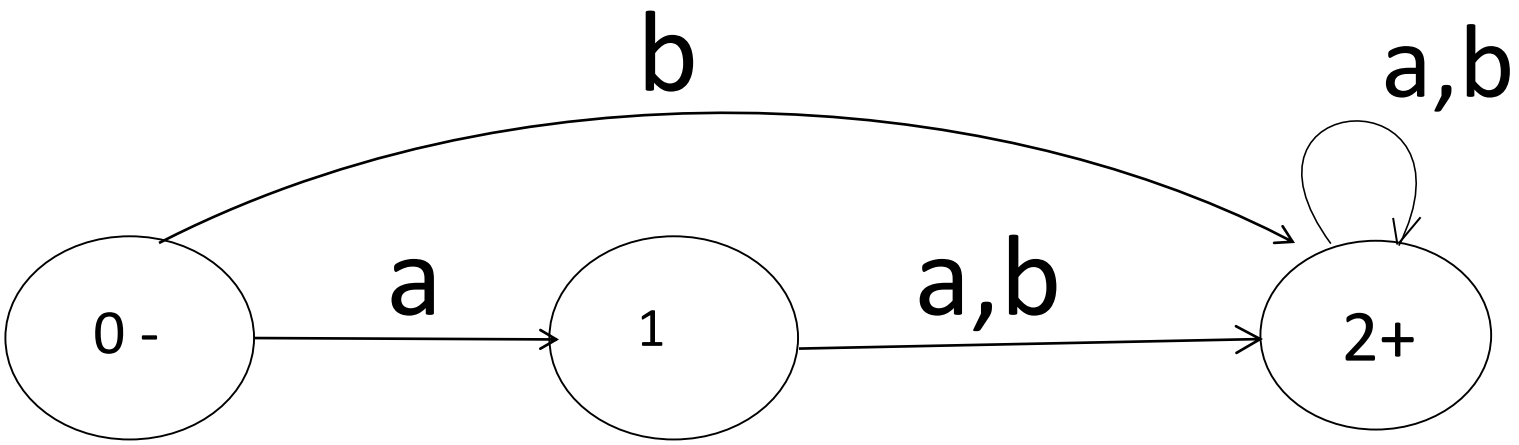
- Σ = {a,b} and states = 0,1,2 where 0 is an initial state and 1 is the final state.
- Transitions:
  1. At state 0 reading a go to state 1,
  2. At state 0 reading b go to state 2,
  3. At state 1 reading a,b go to state 2
  4. At state 2 reading a, b go to state 2

Transition Table Representation

Old state	Input	Next State
0	a	1
0	b	2
1	a	2
1	b	2
2	a	2
2	b	2

Another way of representation…

Old States	New States	
	Reading a	Reading b
0 -	1	2
1	2	2
2 +	2	2


b
a,b
0 -
a
1
a,b
2+